Lecture 5

Haar Measure, Averaging and Orthogonality

Objectives. *For a finite group G, averaging with $|G|^{-1}\sum_{g\in G}$ proves unitarisability, complete reducibility and orthogonality of matrix coefficients. In this lecture we build an integral on a compact Lie group that replaces this average. It is linear and positive, has total mass 1, does not change when the integrand is shifted on either side, and is strictly positive on nonzero nonnegative functions. Sections 5.1 and 5.2 construct it. Sections 5.3 and 5.4 use no other property of it, and they record which property each argument uses.*

By the end of this lecture you will be able to:

- Integrate continuous functions on a compact manifold against a density.
- Construct the normalised Haar integral of a compact Lie group, and prove that it is right invariant and strictly positive.
- Average inner products and operators to prove unitarisability, complete reducibility and Schur orthogonality for finite-dimensional representations of compact Lie groups.
- Compute Haar integrals, an invariant-vector projector and a coefficient norm on $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

Lie groups are real, finite-dimensional, Hausdorff and second countable. Representations are complex, finite-dimensional and continuous. Inner products are antilinear in the first slot.

5.1 Densities and integration on manifolds

Let M be a smooth n-manifold with $n \geq 1$. Charts and tangent spaces are as in Lecture 1. In a chart, the Euclidean integral $\int f\,du$ changes by a Jacobian factor when the chart changes. A density supplies a factor that cancels this change. We cite the routine analytic parts of this construction.

Definition 5.1. Let V be a real vector space of dimension n. A *density* on V is a function $\delta\colon V^n \to \mathbb{R}$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \quad \text{for all linear } T\colon V\to V \text{ and all } v_j \in V.$$

It is *positive* if it is positive on every basis.

A positive density assigns a volume to the box with edges $v_1, \dots, v_n$. The factor $|\det T|$ says how a linear map rescales volume.

Fix a basis $b = (b_j)$ and write $v_j = Tb_j$. Then $\delta(v_1, \dots, v_n) = |\det T|\,\delta(b_1, \dots, b_n)$, so δ is determined by one value. Conversely, let $[v]_b$ be the matrix whose jth column holds the coordinates of v_j. Then

$$\delta_b(v_1, \dots, v_n) = |\det [v]_b|$$

is a density, since $[Tv]_b = [T]_b[v]_b$. Hence the densities on V form the real line $\mathbb{R}\,\delta_b$, and $c\,\delta_b$ is positive exactly when $c > 0$.

In a chart x, let $\partial_{x_j}|_p \in T_pM$ be the velocity of the coordinate curve $t \mapsto x^{-1}(x(p) + te_j)$. These vectors form a basis of T_pM. A tangent vector with coordinate velocity (c_j) is $\sum_j c_j \partial_{x_j}|_p$.

Definition 5.2. A *continuous density* δ on M is a density δ_p on each T_pM such that, for every chart $x\colon U \to \Omega \subset \mathbb{R}^n$, the function ρ_x given by

$$\rho_x(x(p)) = \delta_p\big(\partial_{x_1}|_p, \dots, \partial_{x_n}|_p\big), \qquad p \in U,$$

is continuous on Ω.

Let $y\colon U' \to \Omega'$ be another chart, and put $\phi = y \circ x^{-1}$ on $x(U \cap U')$. The chain rule gives $\partial_{x_j} = \sum_i (\partial y_i/\partial x_j)\,\partial_{y_i}$. The defining property of a density, applied to this change of basis, gives

$$\rho_x(u) = |\det D\phi(u)|\,\rho_y(\phi(u)), \qquad u \in x(U \cap U'). \tag{5.1}$$

Let $f\colon M \to \mathbb{C}$ be continuous and vanish outside a compact subset of U. Define

$$\int_M f\,\delta = \int_\Omega f(x^{-1}(u))\,\rho_x(u)\,du. \tag{5.2}$$

If f vanishes outside a compact subset of $U \cap U'$, then the change-of-variables theorem and the transformation rule (5.1) show that the chart y gives the same value. For example, take spherical angles on S^2, with $p = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, and the density $\delta_p(v_1, v_2) = |\det[\,p\ v_1\ v_2\,]|$. Then the coefficient ρ_x is $\sin\theta$.

Now let M be compact. A finite *partition of unity* is a finite family of continuous functions $\chi_k \geq 0$ with $\sum_k \chi_k = 1$, each vanishing outside a compact subset of a chart domain U_k. We cite the existence of a finite partition of unity on M. For continuous f put

$$\int_M f\,\delta = \sum_k \int_M \chi_k f\,\delta.$$

Let (ψ_l) be a second partition of unity, with chart domains U_l'. Each $\chi_k\psi_l f$ vanishes outside a compact subset of $U_k \cap U_l'$, so its integral is the same in the charts on U_k and on U_l'. Since each partition sums to 1, linearity of the chart integral (5.2) shows that both sums equal $\sum_{k,l} \int_M \chi_k\psi_l f\,\delta$. Hence the integral is well defined. Moreover, it is linear, and it is nonnegative when $f \geq 0$ and δ is positive.

For a diffeomorphism $F\colon M \to M$, the *pullback* density is

$$(F^*\delta)_p(v_1, \dots, v_n) = \delta_{F(p)}(dF_p v_1, \dots, dF_p v_n).$$

The chain rule gives $(F_1 \circ F_2)^* = F_2^* F_1^*$.

Lemma 5.3. *$F^*\delta$ is a continuous density, positive if δ is. For $f \in C(M)$,*

$$\int_M (f \circ F)\, F^*\delta = \int_M f\, \delta.$$

Proof. Let x be a chart on U. Then $x \circ F$ is a chart on $F^{-1}(U)$, and dF sends its coordinate vectors to those of x. So the coefficient of $F^*\delta$ in the chart $x \circ F$ is the continuous function ρ_x. The transformation rule (5.1) then gives continuity in every chart. If f vanishes outside a compact subset of U, both integrals are the same Euclidean integral (5.2). For general f, apply this to each $\chi_k f$ and use the partition of unity $(\chi_k \circ F)$. ∎

In examples we integrate in one chart. Spherical angles on S^2 miss a half great circle, and we ignore it because a curve has zero area. The next lemma states this rule in general. Its proof is cited and uses Lebesgue theory.

One chart often covers M apart from a set contained in finitely many submanifolds of lower dimension. A set $Z \subset \mathbb{R}^n$ is *Lebesgue-null* if, for every $\varepsilon > 0$, countably many boxes of total volume less than ε cover Z. Changing a function on a Lebesgue-null set does not change its Lebesgue integral. We cite the following consequence of Lebesgue theory from Knapp Beyond, Section VIII.1. The integrability statement in the lemma, and the term *measurable* below, belong to that cited theory.

Lemma 5.4. *Let M be compact with a positive continuous density δ. Let $x : U \to \Omega$ be a chart such that $M \setminus U$ lies in a finite union of smooth submanifolds of dimension less than n. Then for $f \in C(M)$ the function $(f \circ x^{-1})\rho_x$ is Lebesgue integrable on Ω, and*

$$\int_M f\, \delta = \int_\Omega f(x^{-1}(u))\, \rho_x(u)\, du.$$

The lemma rests on two cited facts. A submanifold of dimension less than n has Lebesgue-null image in every chart. The change-of-variables formula holds for every nonnegative measurable function. For $f \geq 0$ one applies both facts to each $\chi_k f$, using the transformation rule (5.1), and sums over k. A general $f \in C(M)$ is a linear combination of four nonnegative continuous functions, namely the positive and negative parts of the real and imaginary parts of f.

5.2 The Haar integral

Let G be a compact Lie group with Lie algebra $\mathfrak{g} = T_eG$. If $\dim G = 0$, then G is discrete and compact, hence finite. Until the definition of the Haar integral, Definition 5.6, we assume $n = \dim G \geq 1$. Write $L_h(g) = hg$ and $R_h(g) = gh$. We choose a volume at e and carry it to every point by left translation, as Figure 5.1 shows. So fix a positive density δ_e on $\mathfrak{g}$ and define

$$\delta_g(v_1, \ldots, v_n) = \delta_e(dL_{g^{-1}}v_1, \ldots, dL_{g^{-1}}v_n), \qquad v_j \in T_gG. \tag{5.3}$$

Lemma 5.5. *δ is a positive continuous density, and $L_h^*\delta = \delta$ for all $h \in G$. Every continuous density δ' with $L_h^*\delta' = \delta'$ for all h is a constant multiple of δ.*

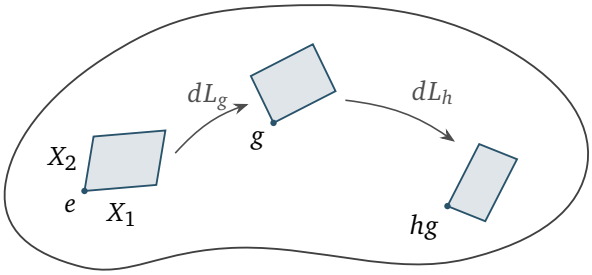


Figure 5.1: The box at e has edges X_1 and X_2. The boxes at g and hg are its images under dL_g and $dL_{hg} = dL_h \circ dL_g$. The density δ takes the same value on all three, although their shapes and areas in a chart may differ.

Proof. Each $dL_{g^{-1}} : T_gG \to \mathfrak{g}$ is a linear isomorphism, so δ_g is positive. Fix a basis (X_j) of $\mathfrak{g}$. The fields $X_j^L(g) = dL_g X_j$ are smooth and form a basis of each tangent space (Lecture 1). In a chart write $\partial_{x_i} = \sum_j a_{ji} X_j^L$. The matrix (a_{ji}) is the inverse of the coefficient matrix of the fields X_j^L in this chart. That coefficient matrix is continuous, hence so is (a_{ji}). Since $dL_{g^{-1}} X_j^L(g) = X_j$, we get $\rho_x = |\det(a_{ji})|\, \delta_e(X_1, \dots, X_n)$, which is continuous. With $v = (v_1, \dots, v_n)$,

$$(L_h^* \delta)_g(v) = \delta_{hg}(dL_h v) = \delta_e(dL_{(hg)^{-1}} dL_h v) = \delta_e(dL_{g^{-1}} v) = \delta_g(v).$$

Now let δ' be left invariant. Taking $h = g^{-1}$ gives $\delta'_g(v) = \delta'_e(dL_{g^{-1}} v)$. The densities on $\mathfrak{g}$ form a real line, so $\delta'_e = c\, \delta_e$ for a real constant c. Hence $\delta' = c\, \delta$. ∎

Definition 5.6. The *normalised Haar integral* of a compact Lie group G is the functional on $C(G)$ given by

$$\int_G f(g)\, dg = \begin{cases} \dfrac{\int_G f\, \delta}{\int_G 1\, \delta}, & \dim G \geq 1, \\[2ex] \dfrac{1}{|G|} \sum_{g \in G} f(g), & G \text{ finite}. \end{cases}$$

The denominator is positive because every ρ_x is positive. Rescaling δ_e does not change the quotient.

Theorem 5.7. *Let G be a compact Lie group, $f \in C(G)$ and $h \in G$.*

(a) The integral is linear, $\int_G f(g)\, dg \geq 0$ if $f \geq 0$, and $\int_G 1\, dg = 1$.
(b) Left invariance: $\int_G f(hg)\, dg = \int_G f(g)\, dg$.
(c) Right invariance: $\int_G f(gh)\, dg = \int_G f(g)\, dg$.
(d) Strict positivity: if $f \geq 0$ and $f(g_0) > 0$ for some g_0, then $\int_G f(g)\, dg > 0$.

Proof. For finite G all parts are clear, since translations permute G. Let $n \geq 1$. Linearity, positivity and normalisation are immediate. Left invariance follows from the pullback lemma, Lemma 5.3, with $F = L_h$, because $L_h^* \delta = \delta$ by the left-invariant density lemma, Lemma 5.5.

(c) By the pullback lemma, Lemma 5.3, $R_h^* \delta$ is a continuous density. Left and right translations commute, so

$$L_k^* R_h^* \delta = (R_h \circ L_k)^* \delta = (L_k \circ R_h)^* \delta = R_h^* L_k^* \delta = R_h^* \delta.$$

By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $R_h^*\delta = c(h)\,\delta$ for a real constant $c(h)$. We compute it at e. The map $L_{h^{-1}} \circ R_h$ is $g \mapsto h^{-1}gh$, whose differential at e is $\mathrm{Ad}(h^{-1})$ (Lecture 2). So for $X = (X_1, \dots, X_n)$ in $\mathfrak{g}$,

$$(R_h^*\delta)_e(X) = \delta_h(dR_h X) = \delta_e(dL_{h^{-1}} dR_h X) = \delta_e(\mathrm{Ad}(h^{-1})X) = j(h)^{-1}\delta_e(X),$$

where $j(h) = |\det \mathrm{Ad}(h)|$. Thus $R_h^*\delta = j(h)^{-1}\delta$. Since Ad is a continuous homomorphism, so is $j\colon G \to (0,\infty)$. Its image is a compact subgroup. If it contained some $t \neq 1$, the powers t^m, $m \in \mathbb{Z}$, would be unbounded. Hence $j = 1$ and $R_h^*\delta = \delta$. Now apply the pullback lemma, Lemma 5.3, with $F = R_h$.

(d) By continuity of f, there exist $c > 0$ and an open set $U \ni g_0$ with $f > c$ on U. The translates aU with $a \in G$ cover G, so compactness (Lecture 3) gives $G = a_1U \cup \cdots \cup a_NU$. Every $g \in G$ has some $a_i^{-1}g \in U$. Since $f > c$ on U and $f \geq 0$ everywhere, we get

$$\Phi(g) = \sum_{i=1}^{N} f(a_i^{-1}g) \geq c \qquad \text{for all } g \in G.$$

By linearity and left invariance, $N\int_G f(g)\,dg = \int_G \Phi(g)\,dg$. By positivity and normalisation, $\int_G \Phi(g)\,dg \geq c > 0$. Hence $\int_G f(g)\,dg > 0$. ∎

We cite uniqueness of the normalised Haar integral from BD I.5. Let $J\colon C(G) \to \mathbb{C}$ be linear with $J(f) \geq 0$ for $f \geq 0$, $J(1) = 1$ and $J(f \circ L_h) = J(f)$ for all $f \in C(G)$ and $h \in G$. Then J is the normalised Haar integral. The left-invariant density lemma, Lemma 5.5, says that the left-invariant density is unique up to a constant factor. This theorem is stronger: it concerns any positive invariant functional on $C(G)$, which is not assumed to come from a density.

We also cite the Riesz representation theorem for positive linear functionals on $C(X)$, with X compact Hausdorff. It shows that the Haar integral is integration against a unique regular Borel probability measure. This is the *normalised Haar measure*. We integrate only continuous functions here, and this lecture does not use this measure. Lecture 9 uses Haar measure for null sets.

5.3 Averaging inner products and operators

Let W be a finite-dimensional complex vector space, and let $F\colon G \to W$ be continuous. We integrate F coordinatewise in a basis of W. The result does not depend on the basis. It satisfies $A\int_G F\,dg = \int_G AF\,dg$ for linear A. It also inherits left and right invariance from the Haar integral, Theorem 5.7.

> **Theorem 5.8** (Unitarisability). *Every representation (Π, V) of a compact Lie group G admits an inner product for which all $\Pi(g)$ are unitary.*

Proof. Choose any inner product $\langle\cdot,\cdot\rangle_0$ on V and set

$$\langle v, w\rangle = \int_G \langle \Pi(g)v, \Pi(g)w\rangle_0\,dg.$$

The integrand is continuous because Π is continuous. By linearity of the integral, the form is antilinear in v and linear in w. It is Hermitian, since the integral commutes with complex

conjugation. For $v \neq 0$, the function $g \mapsto \|\Pi(g)v\|_0^2$ is nonnegative and positive at e. Strict positivity gives $\langle v, v\rangle > 0$. Right invariance, applied to $g \mapsto \langle \Pi(g)v, \Pi(g)w\rangle_0$, gives

$$\langle \Pi(h)v, \Pi(h)w\rangle = \int_G \langle \Pi(gh)v, \Pi(gh)w\rangle_0 \, dg = \langle v, w\rangle. \qquad \blacksquare$$

Compactness cannot be dropped. The boost representation $u \mapsto B(u)$ of $\mathbb{R}$ from Lecture 2 has eigenvalues e^u and e^{-u}, so no inner product makes it unitary. The translation-invariant integral on $\mathbb{R}$ is $\int dt$, and it gives the constant 1 infinite value. The average in the proof is not available.

Corollary 5.9 (Complete reducibility)**.** *Every representation of a compact Lie group is a direct sum of irreducible subrepresentations.*

Proof. Let (Π, V) be a representation, and induct on $\dim V$. If V is irreducible, there is nothing to prove. Otherwise let U be a proper nonzero invariant subspace. Use the invariant inner product of unitarisability, Theorem 5.8. For $u \in U$ and $w \in U^\perp$,

$$\langle u, \Pi(g)w\rangle = \langle \Pi(g)^* u, w\rangle = \langle \Pi(g^{-1})u, w\rangle = 0.$$

So $U^\perp$ is invariant and $V = U \oplus U^\perp$. Apply induction to both summands. $\blacksquare$

The *invariant vectors* of a representation (Π, V) form the subspace

$$V^G = \{v \in V : \Pi(g)v = v \text{ for all } g \in G\}.$$

Proposition 5.10 (Invariant-vector projector)**.** *Let (Π, V) be a representation of a compact Lie group G and $P = \int_G \Pi(g)\, dg$. Then $\Pi(h)P = P = P\Pi(h)$ for all h, and P is a projection with image V^G. If all $\Pi(g)$ are unitary, P is the orthogonal projection onto V^G.*

Proof. Left invariance gives $\Pi(h)P = \int_G \Pi(hg)\, dg = P$. Right invariance gives $P\Pi(h) = P$. The first identity shows $\operatorname{im} P \subset V^G$. For $v \in V^G$ we have $Pv = \int_G v\, dg = v$. Hence $P^2 = P$ and $\operatorname{im} P = V^G$. Now let Π be unitary. As in the proof of complete reducibility, Corollary 5.9, the subspace $U = (V^G)^\perp$ is invariant. Since $\Pi(g)U \subset U$ for all g, also $PU \subset U \cap V^G = 0$. So $U \subset \ker P$. By rank–nullity, $\dim \ker P = \dim V - \dim V^G = \dim U$. Hence $\ker P = (V^G)^\perp$. $\blacksquare$

Let (σ, W) be a second representation. Then $\operatorname{Hom}(V, W)$ carries the representation in which g acts by $E \mapsto \sigma(g)E\Pi(g)^{-1}$. The invariant vectors of this representation are the intertwiners. By the invariant-vector projector, Proposition 5.10, applied to this representation, for every linear $E\colon V \to W$,

$$E^\natural = \int_G \sigma(g)E\Pi(g)^{-1}\, dg \in \operatorname{Hom}_G(V, W). \tag{5.4}$$

5.4 Orthogonality of matrix coefficients

Definition 5.11. Let Π act by unitary operators on a finite-dimensional inner product space V. The *matrix coefficient* of $v, w \in V$ is the function $c^{\Pi}_{v,w}(g) = \langle v, \Pi(g)w\rangle$ on G.

It is antilinear in v and linear in w. Let $D(g)$ be the matrix of $\Pi(g)$ in an orthonormal basis (e_i), so $\Pi(g)e_j = \sum_i D_{ij}(g)e_i$. Then $D_{ij} = c^{\Pi}_{e_i,e_j}$. On $C(G)$ we use

$$\langle f_1, f_2\rangle_{L^2} = \int_G \overline{f_1(g)}\, f_2(g)\, dg.$$

This inner product is also antilinear in the first slot. It is positive definite by strict positivity.

We import Schur's lemma from finite group theory. Its proof is algebraic and holds for any group. For irreducible finite-dimensional complex representations, $\operatorname{Hom}_G(V,W) = 0$ if $V \not\cong W$, and $\operatorname{Hom}_G(V,V) = \mathbb{C}\,\mathrm{id}$.

Theorem 5.12 (Schur orthogonality). *Let (Π, V) and (σ, W) be irreducible unitary representations of a compact Lie group G, with $d = \dim V$.*

(a) If $\Pi \not\cong \sigma$, then $\langle c^{\Pi}_{v,w}, c^{\sigma}_{v',w'}\rangle_{L^2} = 0$ for $v, w \in V$ and $v', w' \in W$.

(b) For $v, w, v', w' \in V$,

$$\int_G \overline{\langle v, \Pi(g)w\rangle}\, \langle v', \Pi(g)w'\rangle\, dg = \frac{1}{d}\, \langle v', v\rangle\, \langle w, w'\rangle.$$

Proof. The key step writes the integrand as a coefficient of an averaged operator. In case (b) put $\sigma = \Pi$ and $W = V$. Define $E\colon V \to W$ by $Eu = \langle w, u\rangle\, w'$. Here $w \in V$ and $w' \in W$, so $E \in \operatorname{Hom}(V,W)$. Since $\Pi(g)^{-1} = \Pi(g)^{*}$,

$$\langle v', \sigma(g) E \Pi(g)^{-1} v\rangle = \langle w, \Pi(g)^{*} v\rangle \langle v', \sigma(g) w'\rangle = \overline{\langle v, \Pi(g)w\rangle}\, \langle v', \sigma(g)w'\rangle.$$

The map $u \mapsto \langle v', u\rangle$ is linear, so it commutes with the integral. Integrating over G therefore gives

$$\int_G \langle v', \sigma(g) E \Pi(g)^{-1} v\rangle\, dg = \langle v', E^{\natural} v\rangle,$$

with the averaged operator $E^{\natural}$ from (5.4). So the left sides in both parts equal $\langle v', E^{\natural} v\rangle$. In case (a), where $\Pi \not\cong \sigma$, Schur's lemma gives $E^{\natural} = 0$. In case (b), where $\sigma = \Pi$, Schur's lemma gives $E^{\natural} = \lambda\,\mathrm{id}$ for some $\lambda \in \mathbb{C}$. Trace is linear and conjugation invariant, so in an orthonormal basis (e_i)

$$d\lambda = \int_G \operatorname{tr}\!\big(\Pi(g) E \Pi(g)^{-1}\big)\, dg = \operatorname{tr} E = \sum_i \langle w, e_i\rangle \langle e_i, w'\rangle = \langle w, w'\rangle.$$

Hence $\langle v', E^{\natural} v\rangle = d^{-1} \langle w, w'\rangle \langle v', v\rangle$. ∎

Corollary 5.13. *Let (Π, V) and (σ, W) be as in Schur orthogonality, Theorem 5.12, with $d = \dim V$. Let D^{Π} and D^{σ} be their matrices in orthonormal bases of V and W. Then*

$$\int_G \overline{D^{\Pi}_{ij}(g)}\, D^{\Pi}_{kl}(g)\, dg = \frac{\delta_{ik}\delta_{jl}}{d}, \qquad \int_G \overline{D^{\Pi}_{ij}(g)}\, D^{\sigma}_{kl}(g)\, dg = 0 \quad (\Pi \not\cong \sigma).$$

In particular $\|c^{\Pi}_{v,w}\|^2_{L^2} = \|v\|^2 \|w\|^2 / d$ for $v, w \in V$.

Proof. Take $v = e_i$, $w = e_j$, $v' = e_k$ and $w' = e_l$ in Schur orthogonality, Theorem 5.12. For the norm identity, take $v' = v$ and $w' = w$ in the relation for a single representation. ∎

In physics, for $G = \mathrm{SU}(2)$ and the module V_m, the entries in an orthonormal weight basis are the Wigner D-functions of spin $m/2$, with $d = m + 1$.

Put $j = i$ and $l = k$, and sum over i and k. The character $\chi_\Pi = \operatorname{tr} \Pi$ is the sum of the diagonal entries D^{Π}_{ii}. Hence the characters of inequivalent irreducible representations are orthonormal. For irreducible unitary Π and σ,

$$\int_G \overline{\chi_\Pi(g)}\, \chi_\sigma(g)\, dg = \begin{cases} 1, & \Pi \cong \sigma, \\ 0, & \Pi \not\cong \sigma. \end{cases}$$

The table records which properties of the integral each argument uses. There, “yes” means that the proof uses the property. Complete reducibility has the same column as unitarisability, since its proof uses only the invariant inner product.

Property	Unitarisability	Projector	Schur orthogonality
Linearity	yes	yes	yes
$\int_G 1\, dg = 1$	only via strict positivity	yes ($Pv = v$)	yes (trace step)
Left invariance	only via strict positivity	yes ($\Pi(h)P = P$)	yes ($E^\natural$ intertwines)
Right invariance	yes	yes ($P\Pi(h) = P$)	no
Strict positivity	yes	no	no

5.5 Examples: the groups U(1) and SU(2)

In each example the single-chart lemma, Lemma 5.4, lets us integrate over one chart.

Example 5.14 (The circle)**.** For $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ use the parametrisation $\theta \mapsto e^{i\theta}$, $0 < \theta < 2\pi$. Its inverse is a chart, whose domain misses only the point 1. At $z = e^{i\theta}$ we have $\partial_\theta = iz$ and $dL_{z^{-1}}(iz) = i$. Taking $\delta_e(i) = 1$ gives $\rho = 1$, so the integral of 1 against δ is 2π. Hence

$$\int_{\mathrm{U}(1)} f(z)\, dz = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta})\, d\theta.$$

The characters $\chi_n(z) = z^n$, $n \in \mathbb{Z}$, are irreducible unitary representations, and

$$\int_{\mathrm{U}(1)} \overline{z^m}\, z^n\, dz = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta}\, d\theta = \delta_{mn}.$$

Example 5.15 (SU(2))**.** Every $g \in \mathrm{SU}(2)$ has the form

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So $g \mapsto (a, b)$ identifies $\mathrm{SU}(2)$ with $S^3 \subset \mathbb{C}^2 = \mathbb{R}^4$. This map and its inverse are restrictions of real linear maps, so the identification is a diffeomorphism. Differentiating $|\omega|^2 = 1$ gives $T_\omega S^3 \subset \omega^\perp$, and both have dimension 3. So $T_\omega S^3 = \omega^\perp$.

The first column of hg is $h(a,b)^T$. Thus L_h acts on S^3 by the unitary matrix h, which is orthogonal on $\mathbb{R}^4$. For $\omega \in S^3$ and $v_j \in \omega^\perp$ set

$$\nu_\omega(v_1,v_2,v_3) = \bigl|\det[\,\omega\ v_1\ v_2\ v_3\,]\bigr|,$$

a real 4×4 determinant. Extending $T\colon \omega^\perp \to \omega^\perp$ by $T\omega = \omega$ shows that ν_ω is a positive density. In a chart with inverse $u \mapsto \omega(u)$, the coefficient of ν is $|\det[\,\omega\ \partial_1\omega\ \partial_2\omega\ \partial_3\omega\,]|$, which is continuous.

For orthogonal A we get $\nu_{A\omega}(Av) = |\det A|\,\nu_\omega(v) = \nu_\omega(v)$. Hence $L_h^*\nu = \nu$. By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $\nu = c\,\delta$ for the left-invariant density δ of (5.3), with $c > 0$. The density ν is the usual volume of the round sphere S^3, and left multiplication rotates this sphere. So the Haar integral on SU(2) is the uniform average over S^3.

Use Hopf coordinates $a = \cos\eta\, e^{i\xi_1}$ and $b = \sin\eta\, e^{i\xi_2}$, with $0 < \eta < \pi/2$ and $0 < \xi_1, \xi_2 < 2\pi$. This parametrisation is injective. The points it misses lie in the circles $\{a = 0\}$ and $\{b = 0\}$ and in the 2-spheres $\{\operatorname{Im} a = 0\}$ and $\{\operatorname{Im} b = 0\}$. In $\mathbb{R}^4$,

$$\begin{aligned}
\omega &= (\cos\eta\cos\xi_1,\ \cos\eta\sin\xi_1,\ \sin\eta\cos\xi_2,\ \sin\eta\sin\xi_2),\\
\partial_\eta\omega &= (-\sin\eta\cos\xi_1,\ -\sin\eta\sin\xi_1,\ \cos\eta\cos\xi_2,\ \cos\eta\sin\xi_2),\\
\partial_{\xi_1}\omega &= (-\cos\eta\sin\xi_1,\ \cos\eta\cos\xi_1,\ 0,\ 0),\\
\partial_{\xi_2}\omega &= (0,\ 0,\ -\sin\eta\sin\xi_2,\ \sin\eta\cos\xi_2).
\end{aligned}$$

These vectors are mutually orthogonal, with lengths $1, 1, \cos\eta, \sin\eta$. So the parametrisation is an injective immersion of an open box. By the inverse function theorem its inverse is a chart. The coefficient of ν in this chart is the product of these lengths, $\cos\eta\sin\eta$. So the integral of 1 against ν is $\frac{1}{2}(2\pi)^2 = 2\pi^2$, the volume of the unit 3-sphere. Therefore

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{2\pi^2}\int_0^{2\pi}\!\!\int_0^{2\pi}\!\!\int_0^{\pi/2} f\,\cos\eta\sin\eta\,d\eta\,d\xi_1\,d\xi_2. \tag{5.5}$$

The monomial $a^p\bar a^q b^r \bar b^s$ carries the phase $e^{i(p-q)\xi_1}e^{i(r-s)\xi_2}$. So its integral vanishes unless $p = q$ and $r = s$. Also

$$\int_{\mathrm{SU}(2)} |a|^2\,dg = 2\int_0^{\pi/2}\cos^3\eta\sin\eta\,d\eta = 2\Bigl[-\frac{\cos^4\eta}{4}\Bigr]_0^{\pi/2} = \frac{1}{2}, \qquad \int_{\mathrm{SU}(2)} |b|^2\,dg = 1 - \frac{1}{2} = \frac{1}{2}.$$

A coefficient norm. The defining representation $V_1 = \mathbb{C}^2$ is irreducible of dimension 2. In the standard basis $D(g) = g$, so $D_{11} = a$ and $D_{12} = -\bar b$. The matrix-entry relations of Corollary 5.13 predict $\int |D_{11}|^2\,dg = \frac{1}{2}$ and $\int \overline{D_{11}}D_{12}\,dg = 0$. Both agree with the direct computation, since $\overline{D_{11}}D_{12} = -\bar a\bar b$ has a nonzero phase.

An invariant-vector projector. Let SU(2) act on $V_1 \otimes V_1$ by $g \otimes g$. Use the orthonormal basis $e_1\otimes e_1$, $e_1\otimes e_2$, $e_2\otimes e_1$, $e_2\otimes e_2$. The entry of $g\otimes g$ in row (i,k) and column (j,l) is $g_{ij}g_{kl}$. The entries are quadratic monomials in $a, \bar a, b, \bar b$. By the vanishing rule for monomials above, only the products $a\bar a$ and $b\bar b$ have nonzero integral. These products occur only in rows and columns $(1,2)$ and $(2,1)$:

$$g_{11}g_{22} = |a|^2, \qquad g_{12}g_{21} = -|b|^2, \qquad g_{21}g_{12} = -|b|^2, \qquad g_{22}g_{11} = |a|^2.$$

Let P be the averaging projector for this action. Writing D for matrices in this basis, we obtain

$$D(P) = \int_{\mathrm{SU}(2)} g \otimes g \, dg = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

With $\varepsilon = e_1 \otimes e_2 - e_2 \otimes e_1$ we get $Px = \frac{1}{2}\langle \varepsilon, x \rangle \, \varepsilon$. This is the orthogonal projection onto $\mathbb{C}\varepsilon$. In physics, ε is proportional to the singlet state of two spin-one-half particles. By the invariant-vector projector, Proposition 5.10, $(V_1 \otimes V_1)^{\mathrm{SU}(2)} = \mathbb{C}\varepsilon$.

Self-checks

1. Which property of the Haar integral makes the form in the proof of unitarisability, Theorem 5.8, positive definite? Which makes it invariant?

   Answer. Strict positivity gives $\langle v, v \rangle > 0$ for $v \neq 0$. Right invariance gives invariance. Left invariance and normalisation enter only through the proof of strict positivity.

2. Let Π be a nontrivial irreducible representation of a compact Lie group. What is $\int_G \Pi(g) \, dg$?

   Answer. It is 0. By the invariant-vector projector, Proposition 5.10, its image is the subrepresentation V^G. Irreducibility gives $V^G = 0$ or $V^G = V$. If $V^G = V$, then every $\Pi(g)$ is the identity, so every line in V is invariant. Irreducibility then forces $\dim V = 1$, so Π is the trivial representation, contrary to the hypothesis. Hence $V^G = 0$, and the integral is 0.

3. Explain $\int_{\mathrm{SU}(2)} a \, dg = 0$ in two ways.

   Answer. First, in Hopf coordinates the integrand carries the phase $e^{i\xi_1}$, so the integral over ξ_1 vanishes. Second, $a = D^{V_1}_{11}$, and the constant 1 is the matrix coefficient of the trivial representation V_0. Since $V_1 \not\cong V_0$, the relation for inequivalent representations in Corollary 5.13 gives $\int_G \overline{1} \, a \, dg = 0$.

Summary

On a compact Lie group G of positive dimension, left translation of one positive density on $\mathfrak{g}$ gives the Haar integral. Compactness forces $|\det \mathrm{Ad}| = 1$, which gives right invariance. A covering argument by translates gives strict positivity. With these properties, the finite-group proofs of unitarisability, complete reducibility and Schur orthogonality carry over with the sum replaced by the integral. By the matrix-entry relations, Corollary 5.13, the matrix entries of an irreducible unitary representation of dimension d in an orthonormal basis are pairwise L^2-orthogonal, and each has squared norm $1/d$. Completeness of matrix coefficients is the Peter–Weyl theorem of Lecture 10.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (BD), Sections I.5 and II.4.

- Sepanski, *Compact Lie Groups*, §1.4, §2.2 and §3.1.
- Knapp, *Lie Groups Beyond an Introduction* (Knapp Beyond), digital second edition, 2023, §VIII.1, for integration and sets of measure zero.